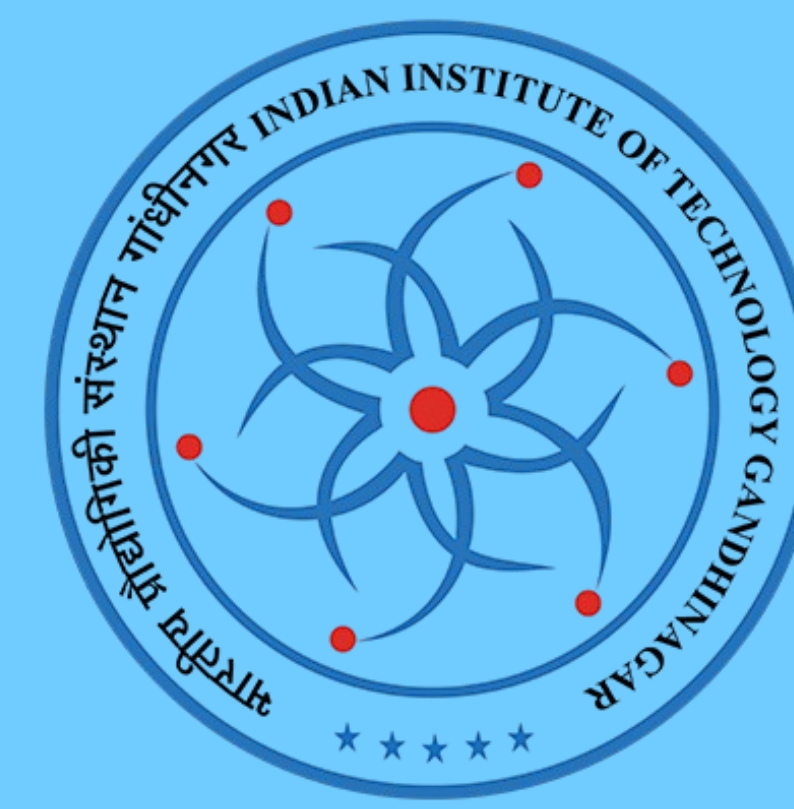


Design and Implementation of an IMU Sensor Network for Real-Time Human Kinematic Reconstruction in Yoga Practice For Elderly



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Motivation and Problem Statement

- **The Home Hazard (User Risk):** Social anxiety and accessibility barriers often limit seniors' ability to engage in unguided home practice (Franco et al., 2015). Lacking professional oversight and proprioceptive awareness, users routinely breach safe biomechanical limits, leading to a 3x higher rate of severe trunk and spinal injuries (Cramer et al., 2013; Swain & McGwin, 2016).
- **The Diagnostic Gap (Clinical Limitation):** Conversely, physical therapists lack accessible and affordable, objective tools to quantify joint angles during dynamic movement. High-fidelity tracking is traditionally restricted to expensive optical camera labs (van der Kruk & Reijne, 2018), leaving routine in-clinic assessments highly subjective.
- **Our Approach:** Engineer a private, low-cost, occlusion-proof wearable system that continuously tracks multi-axis movement, evaluates alignment, and provides real-time biofeedback, and could be used for both personal and clinical usage.

System Application & Architecture

- **Custom Hardware:** A 3-sensor IMU network (Chest, Biceps, Forearms) continuously tracks complex upper-body kinematics and spinal range of motion.
- **Unity Digital Twin:** Real-time 4D Quaternion telemetry drives a responsive 3D avatar. Built-in orthopedic constraints act as a "digital shock absorber," rejecting anatomically impossible angles and enforcing strict safety tolerances (El-Gohary & McNames, 2015; Seel et al., 2014).
- **Clinical Assessment & Data Pipeline:**
 - **Baseline:** Records an expert practitioner creating a perfect "guiding" avatar for exercises like **Tadasana** and the **Back-Scratch test** (Rikli & Jones, 2001).
 - **Step-by-Step Tracking:** Evaluates the user's angular deviation against the guide avatar across 3-4 distinct exercise phases.
 - **Diagnostic Output:** Generates an automated report card, accurately flagging specific joint restrictions based on localized angular discrepancies (Patel et al., 2012; Poitras et al., 2019).
 - **Data Extraction:** Features native OS integration for one-click export of high-fidelity, timestamped clinical datasets (CSV/JSON) for downstream AI processing.

Methodology

- **Hardware Architecture & Sensor Design**
 - **Custom Wearables:** Engineered custom, 3D-printable wearable mounts to securely anchor the sensor network against key anatomical landmarks (Chest, Biceps, Forearms), ensuring zero slip during active assessment (Figures 1 & 2).
 - **Occlusion-Proof Telemetry:** Utilized a network of BNO055 IMUs that rely on gravity and tensor mathematics rather than line-of-sight cameras. This renders the system 100% immune to visual occlusion while strictly preserving patient privacy.
 - **Multiplexed Edge Processing:** Integrated a TCA9548A I²C multiplexer to seamlessly track multiple limbs without bus address conflicts (Figure 3). Data is processed onboard an ESP32 microcontroller and streamed via a high-speed serial pipeline (115200 baud).
- **Kinematic Engine & Software Architecture**
 - **Standalone Clinical Application:** Developed a compiled Unity executable featuring adaptive UI scaling, allowing the software to deploy seamlessly across diverse hardware, from 1080p laptops to 4K clinical displays.
 - **Autonomous Hardware Watchdog:** Engineered a background fail-safe loop that actively monitors the serial data stream. If a Bluetooth or USB interruption occurs, the watchdog autonomously re-establishes the COM port connection without requiring user intervention.
 - **4D Kinematics & Constrained Modeling:** Sensor telemetry is calculated using 4D Quaternion mathematics to prevent Gimbal Lock and mapped to a rigged humanoid mesh (Kuipers, 1999).
 - **Dynamic Calibration (The 'Tare' System):** A programmable neutral-pose calibration step overrides default magnetic alignment, flawlessly locking the patient's physical real-world orientation to the digital space.

Methodology

3D printable designs for wearables

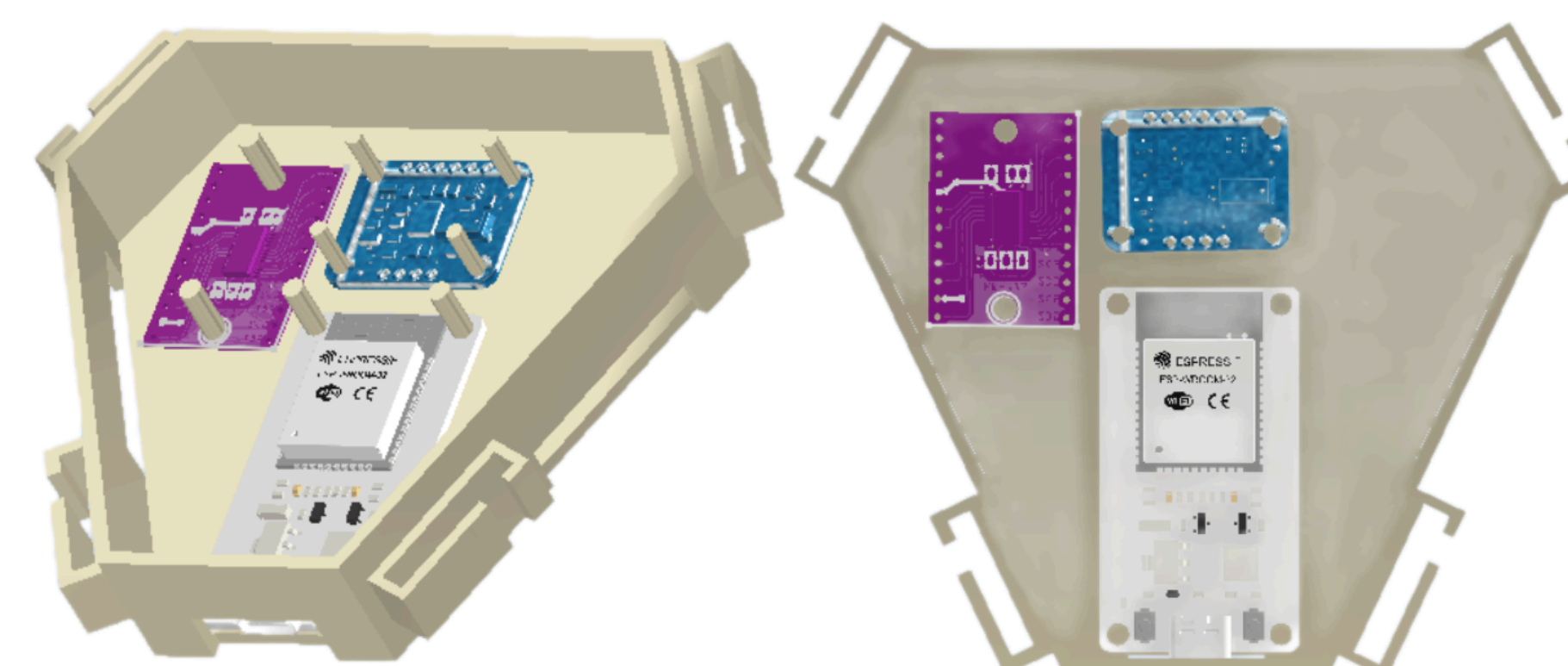


Fig 1 : Design for chest piece.

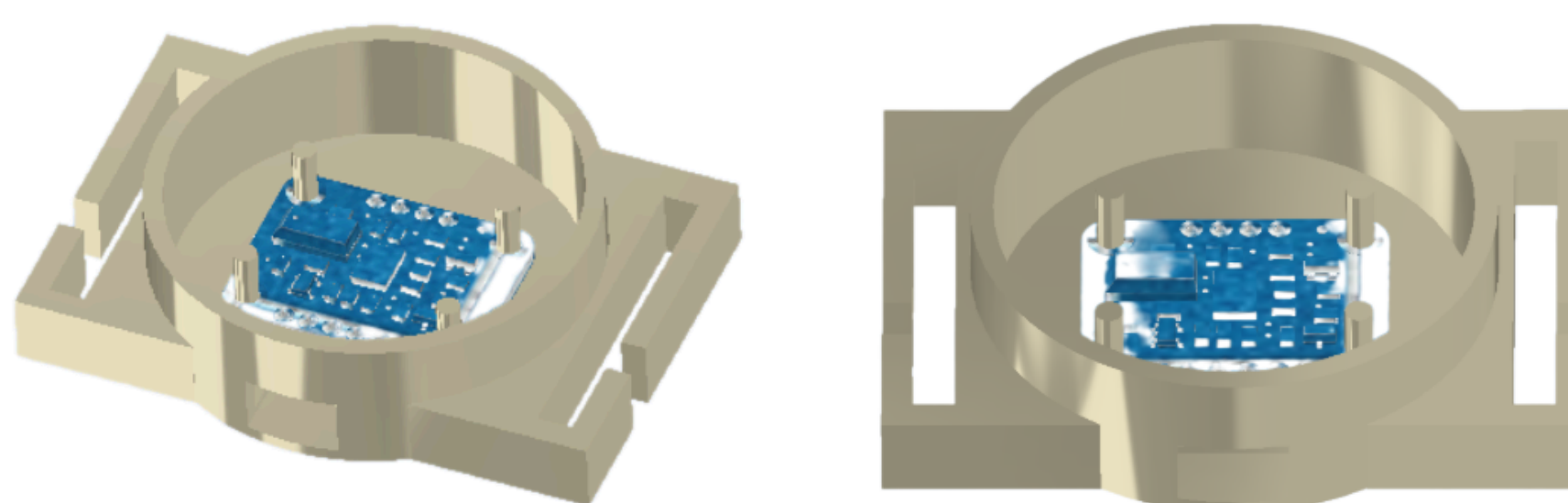


Fig 2 : Design for forearm and bicep piece

Circuit Diagram

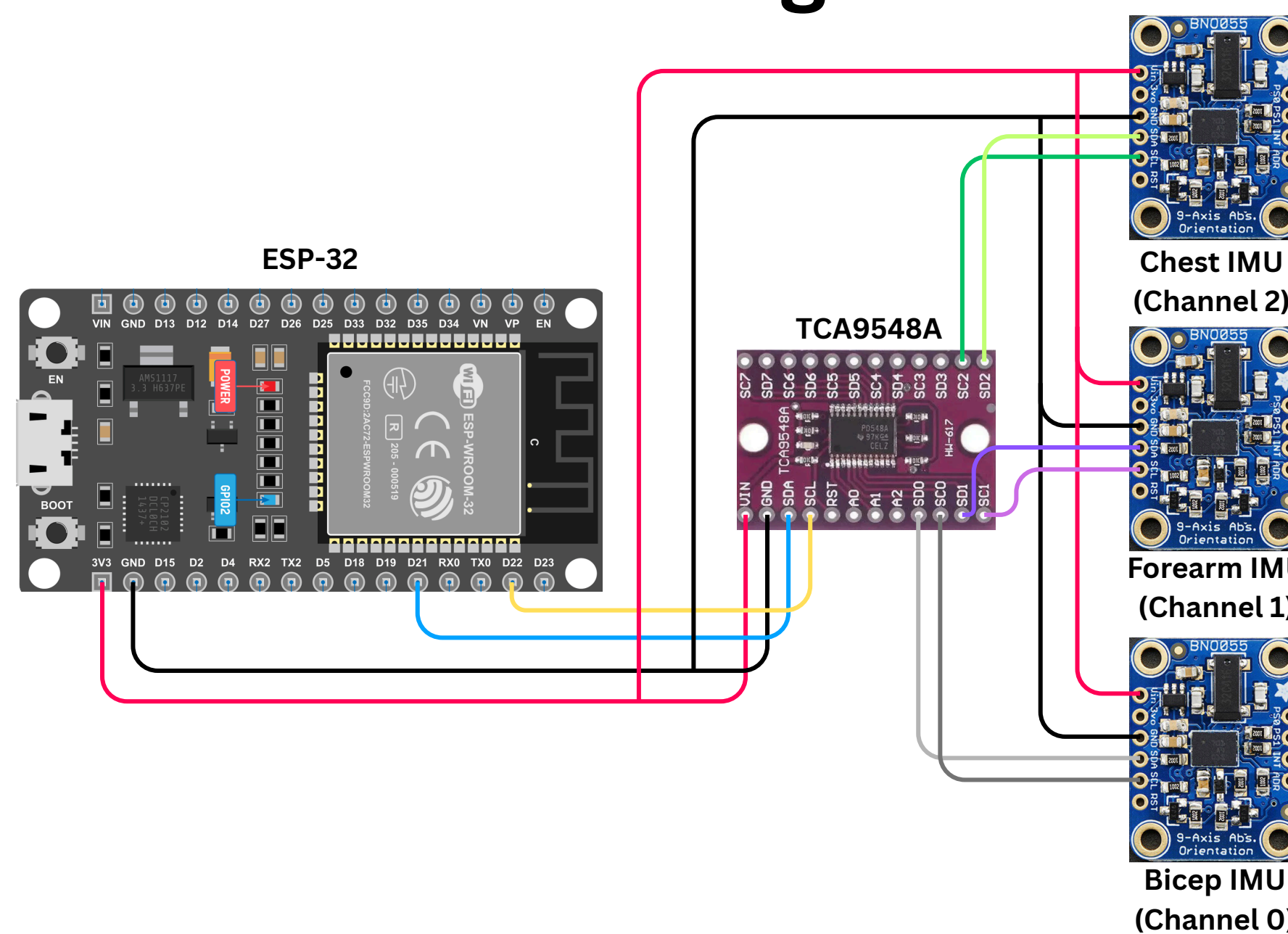


Figure 3 : Circuit representation

- **Clinical Assessment & Data Pipeline**
 - **Phase-Based State Machine:** The software generates a "guide" avatar from baseline recordings of ideal clinical movements. A finite state machine continuously evaluates the routine step-by-step, calculating the precise 3D angular deviation between the live patient and the guide.
 - **Automated Diagnostics:** Upon routine completion, the system logs specific angular deviations across all phases to generate a comprehensive clinical report card, actively flagging potential mobility restrictions or postural deformities.
 - **Native OS Data Extraction:** Bypassing closed-ecosystem limitations, the application features native Windows file browser integration. Clinicians can execute one-click exports of high-fidelity, timestamped kinematic datasets (CSV/JSON), bridging raw physical therapy data with downstream AI analysis platforms.
- **Kinematic Smoothing & Artifact Rejection:**
 - To mitigate inherent IMU sensor noise and transient magnetic interference (Madgwick et al., 2011), the engine employs Spherical Linear Interpolation (SLERP) for all joint rotations (Shoemaker, 1985). Furthermore, an algorithmic transient filter continuously monitors frame-to-frame angular velocity and autonomously rejects anomalous data spikes (glitches) that exceed physiological speed limits, ensuring a fluid, medically accurate digital twin.
- **Optimized Clinical Workflow (UX/UI):**
 - The application architecture utilizes a deterministic Finite State Machine (FSM) to guide clinicians through a standardized assessment loop (Calibration → Preparation → Action → Reporting). The user interface features adaptive, screen-space resolution scaling, ensuring the clinical dashboard remains proportionate and accessible across diverse medical hardware

Results and Discussions

- **Hardware Reliability:** The ESP32 and I²C multiplexer architecture successfully maintained a stable, high-speed telemetry stream (115200 baud) from three independent IMUs.
- **Kinematic Accuracy & Engine Stability:** The Unity 6 digital twin flawlessly interpreted 4D quaternion data with negligible visual latency. The constrained biomechanical model successfully rejected sensor drift and noise, preventing graphical distortion (Gimbal Lock) during complex movements like overhead flexion.
- **Successful Software Deployment:** The system was successfully compiled into a standalone Windows application with adaptive resolution scaling. The native OS integration flawlessly exported structured, timestamped kinematic datasets (CSV/JSON), validating the system's readiness for real-world clinical data extraction.

Future Work

- **Big Data & Predictive AI Diagnostics:** Leverage the system's native CSV/JSON export pipeline to aggregate anonymized kinematic data from thousands of diverse patients. This longitudinal dataset will be used to train predictive machine learning models, allowing the software to autonomously identify the early onset of musculoskeletal deformities and postural diseases before clinical symptoms appear.
- **Gold-Standard Clinical Validation:** Conduct rigorous, IRB-approved comparative trials against industry-standard optical motion capture systems (e.g., Vicon) within real hospital settings. This will statistically quantify the system's angular accuracy and pave the way for formal medical device certification.
- **Hardware Evolution & Smart Garments:** Transition the sensor network from rigid, 3D-printed housings to flexible PCB architectures and conductive textiles. Integrating the IMUs directly into breathable "smart garments" will maximize patient comfort and compliance during extended, unsupervised home monitoring.
- **Immersive Therapeutics (AR/VR):** Expand the Unity digital twin into spatial computing environments. By utilizing Augmented Reality headsets, the system can superimpose the optimal "Ghost Avatar" directly into the patient's physical living room, providing an unparalleled, immersive motor-learning experience.

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